

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

Name	JOANNE CHING YIN XUAN
IC No. / Matric No.	A23CS0227
Year / Program	2/SECP
Section	02
Lecturer Name	DR.SEAH CHOON SEN
Case Study	ADMIN APPROVAL AND GENERATE REPORT
System Name	KOPERASI KAKITANGAN KADA SYSTEM
Group Name	TECHMEDICO

SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description

The Work-Based Learning (WBL) project focuses on enhancing administrative functions in KADA by developing a new system that replaces the outdated manual processes. The current system is dependent on physical forms and manual workflows in membership approval, loan applications, and saving account management. These processes are time-consuming, error-prone, and inefficient, leading to delays and frustrating users. The proposed system will simplify these administrative tasks by introducing an automated, digital approach to increase efficiency and accuracy, hence giving better user satisfaction. With this new system, the administrative module will provide administrators with an easy and efficient way to carry out functions involving loan approvals, membership approvals, saving accounts, and generating annual reports.

The WBL project was initiated because there were several critical issues within the manual system. Firstly, the manual approval processes of memberships, and loans, caused huge inefficiencies and delays. For instance, in most cases, administrators had to manually review and process paper forms which increased the chance of mistakes. Secondly, the lack of an efficient platform meant that there was a communication gap between the administrators and the users, which greatly caused slow responses and decreased user satisfaction. On the other hand, the already existing system had very limited reporting capabilities, so the administrators couldn't easily get insightful data regarding vital information such as membership, loans, and financial transactions. These challenges made it very apparent that a new, updated system that would help automate processes, improve communication, and better decision-making was needed, which prompted the creation of the WBL project. Therefore, my group has proposed a comprehensive system focused on 6 major Modules that include User Authentication Module, Membership Application Module, Loan Application Module, Report Generating Module, Administration Module, and Saving Account Module. I will mainly focus on my module, which is the administrative module in my report. Below is the table that show the module distribution in our group.

Group Members' Name	Matric No	Module
Elijah She Yu Sheng	A23CS0073	Loan Application Module User Authentication Module
Choh Jing Yi	A23CS0296	Report Generating Module
Lee Yin Shen	A23CS0236	Membership Application Module
Joanne Ching Yin Xuan	A23CS0227	Administrative Module
Lim Yu Han	A23CS0241	Saving Account Module

1.2 Project Weaknesses or Improvements

Based on my current module which is an administrative module has several weaknesses that significantly reduce its usability and efficiency, thus making it less effective for both administrators and members. The proposed administrative module, with significant improvement from the manual system, still has several serious weaknesses. One of its weaknesses is its **inability to allow administrators to edit loan rates through a user-friendly interface**. Currently, any adjustment to loan rates requires coding knowledge, making it challenging for administrators without a background in computer science or programming. The reliance on IT personnel for even minor changes prevents timely alterations and creates operational inefficiencies. Furthermore, this reliance restricts the system's capability to adapt quickly to policy changes or market forces, which can negatively impact user satisfaction and the organization's responsiveness.

The second is **any update on the management of loan policy**, for example, eligibility terms for loan applications **require also direct code modifications**. This imposes a strong barrier for non-technical administrators who do not understand coding, thus forcing reliance on developers with each policy change. It causes delays in applying urgent changes, particularly when the transition is to new organizational policies. In addition, constant code changes elevate the likelihood of bringing errors or bugs into the system that can cause interruptions. Part of the solution for this is to have policy content management features that provide an easy text editing window in the admin dashboard to modify policies straight to the user interface.

Lastly, the module's **inability to delete members directly from the admin interface and the dashboard clutter with all applications** is also one of the weaknesses. At present, admins need to go into the MySQL database manually to delete records of members, which is inefficient and error-prone. This limitation clutters the admin dashboard with inactive user applications, making it harder to manage active users effectively. Manual database deletions also increase the chances of partial or incorrect removal of data, thus leaving orphaned records that might affect system performance. Therefore an implementation to allow admins to see all users in one spot, filtered down by criteria, such as approved, rejected, or pending, and the deletion of inactive members from the site is needed. The pending applications will be the default state shown when the admin enters the system. This would greatly facilitate the management of users, lighten the workload of the administrator, and keep the system interface neat.

These identified weaknesses highlight a need for changes that would aim to increase the flexibility and user-friendliness of the system. The administrative module should be redesigned to allow a more intuitive interface that would enable changes to be effected on loan rates, policies, and deletion of applications. Making the system responsive and efficient to the requirements of the organization would only be achieved if the administrators could independently control these important components.

1.3 Proposed Module

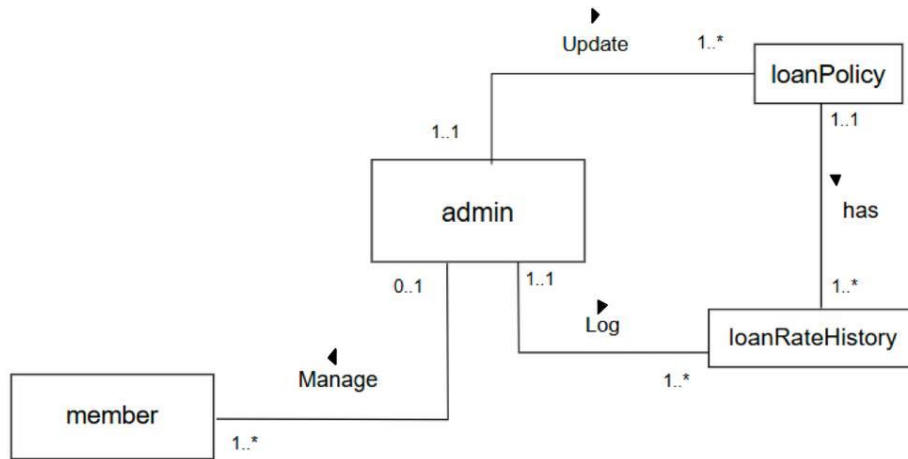
The Enhanced Administrative Dashboard Module is introduced to improve the general system functionality from the identified weaknesses. First, the loan rate adjustment and inefficiency in updating loan policies will be resolved using policy content management. This will add an administrator user-friendly dashboard for loan rate adjustments easily without requiring code changes. The loan rate can be changed directly by the administrator using an input field. This interface will also provide a user-friendly text editor for the modification and publishing of loan eligibility terms, and other details about the policy directly via the interface. This will ease the process and let the administrators make urgent changes in policy without the help of any developer through an easy text editing window in the admin dashboard to modify and publish policies straight to the user interface.

Besides, to cope with the issue of keeping member records, a member management feature will be implemented. This system will have a unified view of all membership records for administrators and will allow them to filter members on various criteria, such as active or inactive. The default view will list active memberships for administrators to take necessary action on those. In addition, the administrators will be able to delete the members straight from the interface if the members want to quit the cooperation or the status is inactive, without having to manually intervene in the database, which is a very error-prone and inefficient task. This will reduce the clutter in the admin dashboard, hence making it easier to manage the active users efficiently. Additionally, deletions in the system will be with checks to ensure data integrity, avoiding orphaned records, thus ensuring the overall performance of the system.

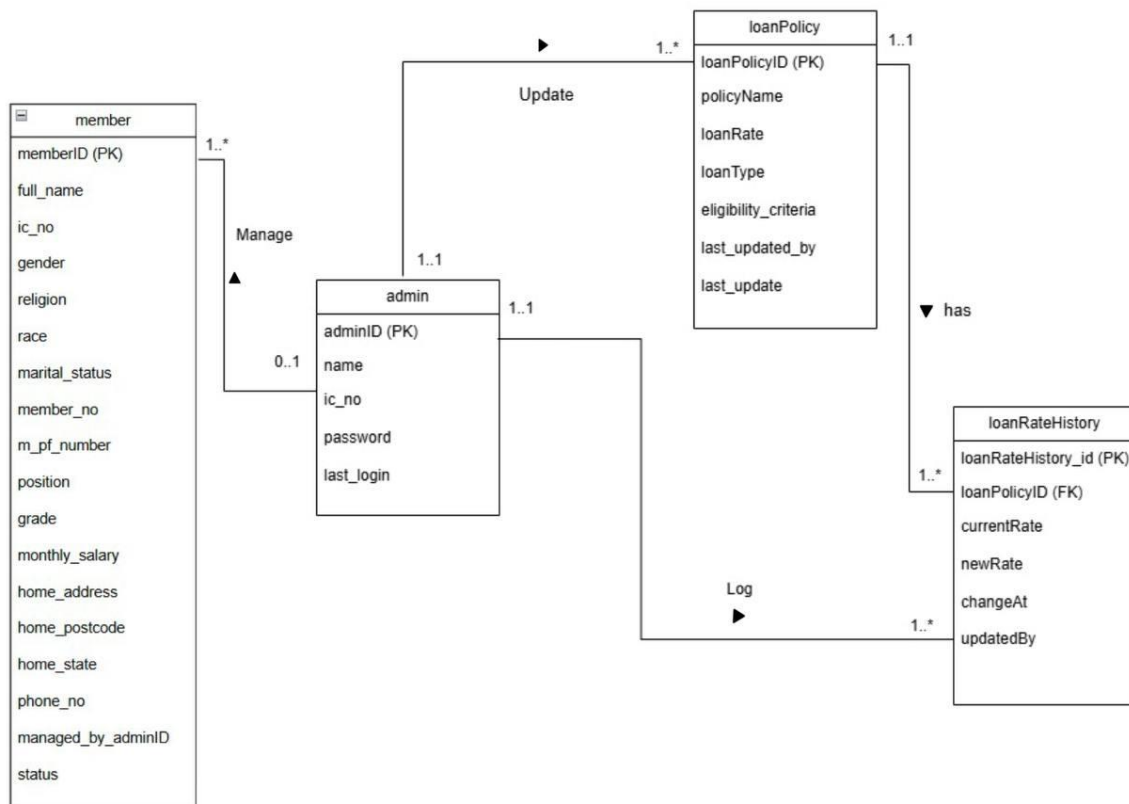
Taken together, these changes being proposed will make the administrative module flexible, efficient, and more user-friendly so that administrators themselves can maintain its key components, such as loan rates, policies, and membership records. In short, by making these processes smooth and eliminating much dependence on IT support, the system will become more agile, and responsive, and meet both the administrators' and members' needs.

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD



2.2 Logical ERD



2.3 Data Dictionary for Logical ERD

Entity: member

Attribute	Data Type &Length	Description	Null
memberID	INT, AUTO_INCREMENT, PRIMARY KEY	Uniquely identifies the member	No
full_name	VARCHAR(225)	Full name of the member	No
ic_no	VARCHAR(20)	Member's Identification Number	No
gender	ENUM('Male', 'Female')	Gender of the member	No
religion	VARCHAR(20)	Religion of the member	No
race	VARCHAR(20)	Race of the member	No
marital_status	ENUM('Single', 'Married', 'Divorced', 'Widowed')	Marital status of the member	No
m_pf_number	VARCHAR(20)	Member Personal File Number	No
position	VARCHAR(100)	Member's job position	No
grade	VARCHAR(20)	Member's employment level	
monthly_salary	DECIMAL(10,2)	Member's monthly salary	No
home_address	TEXT	The residential address of the member	No
home_postcode	VARCHAR(10)	Postal code for the member's home address.	No
home_state	VARCHAR(10)	State of the member's home address.	No
phone_no	VARCHAR(20)	Member's phone number	No
status	ENUM('Active', 'Inactive')	Membership status indicating current active status	No
managed_by_adminID	INT	Foreign key referencing adminID in the admin table.	No

Entity: Admin

Attribute	Data Type &Length	Description	Null
adminID	INT, AUTO_INCREMENT, PRIMARY KEY	Uniquely identifies the admin	No
name	VARCHAR(100)	Full name of the admin	No
ic_no	VARCHAR(20)	The identity card number of the admin	No
password	VARCHAR(50)	Password for admin login	No
last_login	TIMESTAMP	Timestamp of the last login	No

Entity: LoanRateHistory

Attribute	Data Type &Length	Description	Null
loanRateHistory_id	INT, AUTO_INCREMENT, PRIMARY KEY	Uniquely identifies the admin	No
loanPolicyID	VARCHAR(100) FOREIGN KEY	Link to the unique identifier for the loan policy of loanPolicy table	No
currentRate	DECIMAL(5, 2)	Current rate for the loan	No
newRate	DECIMAL(5, 2)	New rate for the loan policy	No
changeaAt	TIMESTAMP	Timestamp of the last change	No
updatedBy	DATETIME	Admin ID of the person who last updated the loan rate	No

Entity: LoanRateHistory**Entity: loanPolicy**

Attribute	Data Type &Length	Description	Null
loanPolicyID	INT, AUTO_INCREMENT, PRIMARY KEY	Unique identifier for the loan policy	No
policyName	VARCHAR(50)	Name of the policy	No
loanRate	DECIMAL(5, 2)	The rate of the loan	No
loanType	VARCHAR(20)	The loan type which have Pembiayaan_Skim_Khas, Pembiayaan_Al_Bai, Pembiayaan_RoadTaxInsuran, Pembiayaan_Al_Innah	No
eligibility_criteria	TEXT	Eligibility criteria for the loan	No
last_update	TIMESTAMP	Timestamp of the last update	No
last_updated_by	INT	Admin ID of the person who last updated the policy	No

2.4 Relational Database Schema

Entity 1	Relationship	Entity 2	Multiplicity
admin	Update	loanPolicy	1...*
admin	Manage	member	1...*
admin	Log	loanRateHistory	1...*
loanPolicy	Has	loanRateHistory	1...*

Kada_database (relational database schema)

member(member_id, full_name, ic_no, gender, religion, race, marital_status, m_pf_number, position, grade, monthly_salary, home_address, home_postcode, home_state, phone_no, status, managed_by_adminID)

PK: member_id

FK: managed_by_adminID REFERENCES admin(admin_id)

admin(adminID, name, ic_no, password, last_login)

PK: adminID

loanPolicy(loanPolicyID, policyName, loanRate, loanType, eligibility_criteria, last_update, last_update_by)

PK: loanPolicyID

FK: last_update_by REFERENCE admin(adminID)

loanRateHistory

(loanRateHistory_id, loanPolicyID, currentRate, newRate, changeAt, updatedBy)

PK: loanRateHistory_id

FK: loanPolicyID REFERENCE loanPolicy(loanPolicyID)

FK: updatedBy REFERENCES admin(adminID)

2.5 Normalization

member table

1NF

member(member_id, full_name, ic_no, gender, religion, race, marital_status, m_pf_number, position, grade, monthly_salary, home_address, home_postcode, home_state, phone_no, status). The Member table satisfies 1NF as all attributes have no repeating groups.

Primary Key: member_id

2NF

member_id	full_name	ic_no	gender	religion	race	marital_status	home_address	home_postcode	Home_state	managed_by_adminID
1	JOANNE CHING	740604010122	Female	Buddhism	Chinese	Single	18, JALAN KEBANGSAAN 3, TAMAN UNIVERSITI	81300	Johor	1
2	CALYN LIM	740604010122	Female	Christianity	Chinese	Married	26, JALAN INDAH 14, TAMAN MUTIARA	81200	Johor	2

Primary Key: member_id

m_pf_number	position	grade
MPF123	accountant	51
MPF123	executive	53

Primary Key: m_pf_number

3NF

Since there are no partial dependencies in the relation, all derived relations are already in third normal form (3NF). Therefore, all derived dependencies will remain the same, as in the second normal form (2NF).

member_id	full_name	ic_no	gender	religion	race	marital_status	home_address	home_postcode	Home_state	managed_by_adminID
1	JOANNE CHING	740604010122	Female	Buddhism	Chinese	Single	18,JALAN KEBANGSAAN 3,TAMANN UNIVERSITY	81300	Johor	1
2	CALYN LIM	740604010122	Female	Christianity	Chinese	Married	26,JALAN INDAH 14,TAMANN MUTIARA	81200	Johor	2

Primary Key: member_id

m_pf_number	position	grade
MPF123	accountant	51
MPF123	executive	53

Primary Key: m_pf_number

admin table

1NF

Each column contains no multi-valued fields or repeated groups

adminID	name	ic_no	password	last_login
1	Zuraini binti Abdullah	740604010123	Zuraini@1112	2025-01-23 10:45:00
2	Azmi bin Muhammad	800714020666	AzmiMuh@12356	2025-01-23 15:30:00

Primary Key:adminID

2NF

adminID is the only primary key, and all attributes (name, ic_no, password, last_login) fully depend on adminID. There are no partial dependencies. Thus, the table is already in 2NF.

adminID	name	ic_no	password	last_login
1	Zuraini binti Abdullah	740604010123	Zuraini@1112	2025-01-23 10:45:00
2	Azmi bin Muhammad	800714020666	AzmiMuh@12356	2025-01-23 15:30:00

Primary Key:adminID

3NF

All attributes are directly dependent on adminID and there are no transitive dependencies. Thus, the table is already in 3NF

adminID	name	ic_no	password	last_login
1	Zuraini binti Abdullah	740604010123	Zuraini@1112	2025-01-23 10:45:00
2	Azmi bin Muhammad	800714020666	AzmiMuh@12356	2025-01-23 15:30:00

Primary Key:adminID

The original table statisfies the 1NF,2NF ,and 3NF.So no further normalization needed.

loanPolicy table

1NF

Each column contains no multi-valued fields or repeated groups

loanPolicy ID	policyName	loan Rate	loanType	eligibility_criteria	last_update	last_update_by
1	Policy of Pembiayaan_Al_Innah	4.20 %	Pembiayaan_Al_Innah	Only available for cooperative"s employees	2025-01-22 14:00	1
2	Policy of Pembiayaan_Skim_Khas	4.20 %	Pembiayaan_Skim_Khas	Only available for cooperative"s employees	2025-01-22 14:30	2

Primary Key:loanPolicyID

2NF

loanPolicyID is the only primary key, and all attributes (policyName, loanRate, loanType, eligibility_criteria, last_update, last_update_by) fully depend on loanPolicyID. There are no partial dependencies. Thus, the table is already in 2NF.

loanPolicy ID	policyName	loan Rate	loanType	eligibility_criteria	last_update	last_update_by
1	Policy of Pembiayaan_Al_Innah	4.20 %	Pembiayaan_Al_Innah	Only available for cooperative"s employees	2025-01-22 14:00	1
2	Policy of Pembiayaan_Skim_Khas	4.20 %	Pembiayaan_Skim_Khas	Only available for cooperative"s employees	2025-01-22 14:30	2

Primary Key:loanPolicyID

3NF

All non-prime attributes (policyName, loanRate, loanType, eligibility_criteria, last_update, last_update_by) depend directly on the primary key loanPolicyID. There are no transitive dependencies where a non-prime attribute depends on another non-prime attribute. Thus, the table is already in 3NF.

loanPolicyID	policyName	loanRate	loanType	eligibility_criteria	last_update	last_update_by
1	Policy of Pembiayaan_Al_Innah	4.20 %	Pembiayaan_Al_Innah	Only available for cooperative's employees	2025-01-22 14:00	1
2	Policy of Pembiayaan_Skim_Khas	4.20 %	Pembiayaan_Skim_Khas	Only available for cooperative's employees	2025-01-22 14:30	2

Primary Key: loanPolicyID

The original table satisfies the 1NF, 2NF, and 3NF. So no further normalization needed.

loanRateHistory table

1NF

Each column contains no multi-valued fields or repeated groups

loanRateHistory_id	loanPolicyID	currentRate	newRate	changeAt	updatedBy
1	1	4.00%	4.20%	2025-01-01 10:30	1
2	2	4.00%	4.20%	2025-01-01 10:30	2

Primary Key:loanRateHistory_id

2NF

loanRateHistory_id is the only primary key, and all attributes (loanPolicyID, currentRate, newRate, changeAt, updatedBy) fully depend on loanRateHistory_id. There are no partial dependencies. Thus, the table is already in 2NF.

loanRateHistory_id	loanPolicyID	currentRate	newRate	changeAt	updatedBy
1	1	4.00%	4.20%	2025-01-01 10:30	1
2	2	4.00%	4.20%	2025-01-01 10:30	2

Primary Key:loanRateHistory_id

3NF

All non-prime attributes (loanPolicyID, currentRate, newRate, changeAt, updatedBy) depend directly on the primary key loanRateHistory_id. loanPolicyID and updatedBy are foreign keys, meaning they refer to data in other tables (loanPolicy and admin table). They do not depend on other non-key attributes. So, it shows that there are no transitive dependencies where a non-prime attribute depends on another non-prime attribute. Thus, the table is already in 3NF.

loanRateHistory_id	loanPolicyID	currentRate	newRate	changeAt	updatedBy
1	1	4.00%	4.20%	2025-01-01 10:30	1
2	2	4.00%	4.20%	2025-01-01 10:30	2

Primary Key:loanRateHistory_id

The original table satisfies the 1NF, 2NF, and 3NF. So no further normalization needed.

Final normalized relations achieved

member table

member_id	full_name	ic_no	gender	religion	race	marital_status	home_address	home_postcode	Home_state	managed_by_adminID
1	JOANNE CHING	740604010122	Female	Buddhism	Chinese	Single	18,JALAN KEBANGSAAN 3,TAMANN UNIVERSITY	81300	Johor	1
2	CALYN LIM	740604010122	Female	Christianity	Chinese	Married	26,JALAN INDAH 14,TAMANN MUTIARA	81200	Johor	2

Primary Key: member_id

m_pf_number	position	grade
MPF123	accountant	51
MPF123	executive	53

Primary Key: m_pf_number

admin table

adminID	name	ic_no	password	last_login
1	Zuraini binti Abdullah	740604010123	Zuraini@1112	2025-01-23 10:45:00
2	Azmi bin Muhammad	800714020666	AzmiMuh@12356	2025-01-23 15:30:00

Primary Key: adminID

loanPolicy table

loanPolicy ID	policyName	loan Rate	loanType	eligibility_criteria	last_update	last_update_by
1	Policy of Pembiayaan_Al_Innah	4.20 %	Pembiayaan_Al_Innah	Only available for cooperative"s employees	2025-01-22 14:00	1
2	Policy of Pembiayaan_Skim_Khas	4.20 %	Pembiayaan_Skim_Khas	Only available for cooperative"s employees	2025-01-22 14:30	2

Primary Key:loanPolicyID

loanRateHistory table

loanRateHistory_id	loanPolicyID	currentRate	newRate	changeAt	updatedBy
1	1	4.00%	4.20%	2025-01-01 10:30	1
2	2	4.00%	4.20%	2025-01-01 10:30	2

Primary Key:loanRateHistory_id